

RF-MAF-AttnGRU: A novel hybrid machine learning model for stock price prediction and algorithmic trading

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Abstract

Accurate forecasting of financial market prices helps investors to make financial decisions and even the government to formulate appropriate macroeconomic policies in a country. In this work, we propose a hybrid model called RF-MAF-AttnGRU, combining random forest (RF), moving average filter (MAF), and gated recurrent units (GRUs) based on an attention mechanism. Firstly, input features such as historical data and technical indicators of the stock market are ranked in importance by RF. Secondly, a MAF based decomposition procedure is developed to decompose the filtered feature time series into the trend and residual series, which can avoid the data leakage problem. Then, the GRU based on the attention mechanism is used to model these two series for the price prediction task. Six stocks listed in China and the US were used to evaluate the performance of the proposed model. The prediction results of the RF-MAF-AttnGRU are compared with five classical prediction models and four state-of-art time series prediction models. The experimental results show that the proposed model yields the best results on all four evaluation metrics among all the algorithms. To further demonstrate the usefulness of the prediction model, we develop a corresponding trading strategy based on the price prediction results of the proposed model. Then, the trading strategy is compared with other trading strategies in terms of common return indicators. The annualized return of the trading strategy based on the proposed model is 23.08%, which is much higher than the best-annualized return of the other trading strategies with only 9.37%.

Keywords: Random forest, Gated recurrent unit, Financial Time Series Forecasting, Attention mechanism, Algorithmic trading

1. Introduction

The stock market is considered to be an economic barometer of a country or region [1]. Accurate stock market forecasts not only provide effective decision-making for investors but also provide direction for the government in establishing appropriate and credible economic

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policies. Therefore, forecasting stock prices is of great significance to policymakers and investors. Hence, stock market forecasting, a classical problem, has attracted researchers from a wide range of fields. However, fluctuations in share prices are always influenced by various factors, such as national policies, global economic developments, and the investment strategies of financial organizations. Consequently, financial market predictions always fail to achieve the expected results.

In the early stages of research, econometric methods are usually adopted for stock price forecasting. Researchers apply traditional statistical models to forecast stock price trends and guide the investment activities of investors. The commonly deployed models are Exponential Smoothing (ES), Autoregressive Integrated Moving Average (ARIMA) [2], and Autoregressive Conditional Heteroskedasticity (GRACH) [3]. However, econometric models have difficulty in dealing with nonlinear systems and require stringent assumptions. These limitations make them less effective in predicting nonlinear, non-stationary and highly noisy financial time series [4, 5].

Compared to statistical models, machine learning methods can explore the nonlinear relationships between target data without setting any assumptions. Researchers then tried to predict financial time series through machine learning methods. Cavalcante et al. found that deep learning models with nonlinear mapping capabilities can accurately predict financial time series trends [6]. In addition, attention mechanisms can greatly improve the performance of neural network models and have been widely used in the fields of machine translation and image captioning [7]. Hence, researchers tried to explore adopting attention mechanisms as part of a neural network to improve the prediction accuracy of financial time series. Although machine learning models outperform econometric models in dealing with financial time series, there are specific scenarios in which machine learning models are subject to local optimization, over-fitting, and parameter sensitivity problems.

To further improve the predictive capability of financial forecasting methods, researchers established hybrid frameworks by adding various modules. For instance, the researchers use feature selection modules to improve the performance of stock prediction [8]. Selecting the optimal combination of features by discarding redundant or irrelevant features allows the learning algorithm to train the model more efficiently. On the other hand, finding a proper decomposition algorithm to deal with nonlinear time series often improves the predictive performance of the model [9]. Niu et al. decomposed the stock market index by variational mode decomposition (VMD) and used Long short-term memory (LSTM) to forecast the decomposed sub-series. Then, the sub-series output was integrated to generate the final prediction [10]. Although the divide-and-conquer approach can effectively improve the predictive performance of a model, some studies based on time series decomposition algorithms may involve future data in decomposing the test dataset, which can cause data leakage problems [10].

The majority of the previous literature has focused on proposing models that can accurately predict stock market movements. The comparisons of the performance of different methods are always based on traditional regression or classification metrics, which have limitations. Note that one of the critical issues for developing an accurate predictive model is to help investors to make financial decisions, namely, providing complementary trading

strategies to generate profits for investors. However, few of the existing literature provides investors with trading strategies based on prediction results [8]. The few studies that have developed trading strategies through predictive methods also tend to ignore commissions, which results in exaggerated returns [11]. It is worth noting that there are many trading details to consider for the design of a stock trading system based on a predictive model. If the trading strategy is designed improperly, the advantages of a trading decision system based on artificial intelligence might be undermined [12]. Therefore, few studies have tried to build a framework that allows accurate forecasts and provides investors with financial decisions through suitable trading strategies.

For the above challenges, we propose a hybrid model, combining RF, MAF, and GRUs based on attention mechanisms, called RF-MAF-AttnGRU. Firstly, RF is used to evaluate the importance of input features, including historical data and technical indicators of the stock market and to filter out the optimal subset of features. As the decision tree (DT) is constructed in a way that the optimal features are selected to divide the sample data by information gain ratio or Gini index, RF can be utilized to derive the importance score of each feature in the process of stock movement prediction. Additionally, RF is more explanatory than other feature selection methods[13]. Secondly, a series decomposition module based on moving averages is developed to divide the time series into trending and residual series to ensure no data leakage during the decomposition process. Finally, an attention-based GRU was used to forecast the movement of the two sequences for the price prediction task.

Six stocks listed in China or the US are adopted to evaluate the performance of the proposed methods. We compare the proposed model with classical machine learning and state-of-the-art time series prediction models. The experimental results show that RF-MAF-AttnGRU achieves the most accurate results in the forecasting task. In addition, we set up trading strategies based on the prediction results of the proposed model. A commercial trading scenario with the consideration of transaction commission is further settled. Finally, in this scenario, trading strategies based on the proposed model are compared with other traditional trading strategies on common return indicators. The backtest results show that the proposed model is the most desirable in terms of both application prospects and the trade-off between risk and return. The main contributions of this paper are summarized as follows:

- We propose a hybrid framework, called RF-MAF-AttnGRU, that combines a feature selection technique, time series decomposition algorithm, and gated recurrent units based on attention mechanisms. This model has shown superior performance in the prediction of stock prices.
- We compare the forecast performance of the proposed model with classical machine learning and state-of-the-art time series forecasting models on six stocks from various regions. Moreover, we verify the validity of each module in the proposed model by ablation experiments.
- We propose a comparative framework based on nonparametric tests to analyze the composite performance of the regression models on the six stock datasets. The non-

parametric Friedman and Iman-Davenport tests are adopted further to evaluate the comprehensive performance of the various forecasting models.

- A trading strategy, based on the prediction results of the proposed model, is designed and compared with traditional trading strategies in terms of return capability. Experimental results indicate that investors can gain stable returns according to trading strategies based on the proposed model.

The rest of the paper is organized as follows: Section 2 reviews the previous research relevant to our work. Section 3 introduces the theoretical concept design of the model in detail. Then, in Section 4, the experimental design is given. The experiment results of the model are assigned to Section 5. The conclusion and future work are presented in Section 6.

2. Relate Work

Researchers in the stock market widely use deep learning because of its remarkable nonlinear mapping capabilities. For example, Nelson et al. used LSTM to predict stock price trends and developed a trading strategy to obtain higher returns than a buy-and-hold strategy [11]. Dutta concluded that LSTM and GRU outperformed other traditional machine learning models. GRU, with fewer learning parameters, has better prediction performance than LSTM for bitcoin price prediction. Overall, the improved recurrent neural network was suitable for financial time series trend forecasting [14].

The introduction of the attention mechanism into the encoder-decoder network has shown outstanding results in machine translation and image captioning [15]. Hence, researchers have tried to use the attention mechanism as part of a neural network to improve the prediction accuracy of financial time series. Xiao et al. presented a deep neural network based on two-stage attention. The model eliminated the interference of irrelevant information through two stages of attention [16]. Du et al. developed a hybrid framework to predict five classical time series by combining an encoder-decoder based on an attention mechanism and Bi-directional Long Short-Term Memory (Bi-LSTM). Prediction accuracy can also be improved by adaptively learning the long-term correlation of multi-dimensional temporal data and hidden correlation features [17].

As not all features are highly related to the target variable, improvements in model prediction performance can be achieved through suitable feature selection methods. Binoy et al. proposed a method based on DT and Support Vector Machine (SVM) to predict future price trends. In this case, the selection of input features was conducted during DT construction [18]. Nti et al. selected features that influence the stock prices of listed companies from the macroeconomic variables of a country through RF and input them into the LSTM. Experiments at the Ghana Stock Exchange showed that RFs extracted the optimal combination of features affecting the stock price, which achieved an accurate prediction of market movements [19]. Therefore, a suitable feature selection method can effectively filter redundant or irrelevant features and achieve more accurate stock market forecasts.

Recent literature has shown that suitable decomposition algorithms can improve the prediction accuracy of models. Due to the advantages of decomposing nonlinear and non-stationary time series, decomposition algorithms like empirical mode decomposition (EMD), VMD and wavelet transform (WT) are widely used in time series forecasting tasks. Experimental results in several studies have shown that the EMD decomposition module based on the idea of divide-and-conquer can improve prediction performance [20]. According to the idea of Hyndman, the time series can be decomposed into long-term trends, seasonal variations, and cyclical fluctuations [21]. However, the time series decomposition algorithms in some studies involve the use of future data in decomposing the test dataset, which causes problems of data leakage. In the real-life stock forecasting scenario, researchers try to decompose and model the trend series in the financial time series using moving averages to obtain accurate forecasts. Babu used the moving average filter to decompose the time series into linear and nonlinear variation series. ARIMA and ANN were used to model the two sequences separately and integrate the final output. The hybrid model showed the best performance on both electricity price data and stock market data [22]. Moving average filters can avoid the problem of data leakage while obtaining smooth time subseries [9].

The study of financial time series forecasting is not only concerned with constructing models that accurately predict stock market price movements, but also with providing investors with trading strategies to realize profits on investment. Cavalcante et al. summarized the literature on stock market movement forecasting between 2009 and 2015 and proposed a framework for intelligent stock forecasting based on machine learning. This framework emphasized trading strategy formulation and return evaluation besides model construction [6]. Martinez et al. developed a trading system based on ANN forecasting to provide investors with the best timing for trading. Experimental results showed that ANN achieved the best performance in both the classical valuation metric MAPE and the traditional return metric annualised return (AR) [23]. Nelson built LSTM networks to predict future trends in stock prices based on historical prices and technical indicators. A trading strategy based on the model prediction results was designed to conduct trading operations on five stocks in the Brazilian stock market. The experimental results showed that trading strategies based on LSTM could still achieve substantial returns with minimal risk taken [11].

3. Methodology

The closing price of stock Y is settled as the target series. Then, set $\mathbf{X} = (\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^d) = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)^\top \in \mathbb{R}^{T \times d}$ as the input sequence, where T is the total time length and d stands for the number of features. We use $\mathbf{x}^k$ to indicate the time series of the k th feature, while $\mathbf{x}_t$ represents the sequence of features at time t . For instance, $\mathbf{x}^1$ denotes the closing price, $\mathbf{x}^2$ denotes the highest price, $\mathbf{x}^3$ denotes the lowest price, etc. The individual features are summarized in Table 1.

We aim to use historical data with a window size of s to predict the closing price of a stock for the next day. Here, $\hat{\mathbf{Y}} = (\hat{y}_{s+1}, \hat{y}_{s+2}, \dots, \hat{y}_{T+1}) \in \mathbb{R}^{T-s+1}$ represents the sequence of closing price prediction values. Given $\mathbf{X}_s = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_s)^\top \in \mathbb{R}^{s \times d}$ and $\hat{y}_{s+1}$, we have:

$$\hat{y}_{s+1} = F(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_s), \quad (1)$$

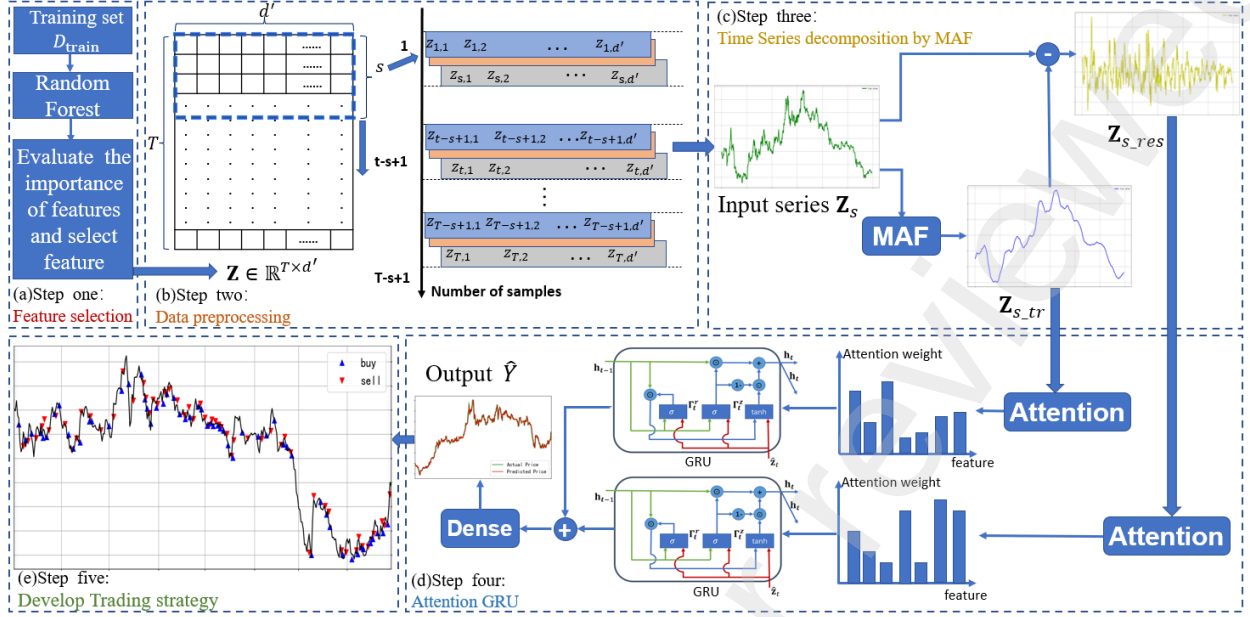


Figure 1: The Flowchart of the stock forecasting framework RF-MAF-AttnGRU; (a) Feature selection by Decision Tree; (b) Data preprocessing to create input time series; (c) Time series decomposition by Moving average Filter; (d) Forecast price Attention GRU; (e) Develop trading strategy and quantify backtest trading.

where $F(\cdot)$ is a nonlinear mapping function we aim to learn from the historical data.

3.1. Overview of the Proposed Design

In this work, we propose the RF-MAF-AttnGRU framework process for stock prediction. The proposed RF-MAF-AttnGRU has five main stages: 1) RF is used to rank the importance of the historical data and technical indicators and to filter out the optimal subset of features; 2) The Sliding window method and the maximum-minimum normalization technique are utilized to pre-process the stock data to generate the input time series; 3) MAF is adopted to decompose the selected time series into trend and residual series; 4) AttnGRU, the gated recurrent units based on the attention mechanism, is deployed to model these two sequences for the price prediction task; 5) We set up trading strategies based on the prediction results of the proposed model. The trading strategies are determined based on the relationship between the predicted closing price of the day ahead and the true closing price of the day. The framework process for stock forecasting is shown in Figure 1. Next, each step will be described in detail.

3.2. Random Forest

Random forest (RF) is a classical ensemble learning model composed of multiple DTs [24]. In the complex stock market, each feature has a different degree of influence on the price. Since DT is constructed in a way that the optimal features are selected to divide the sample data by methods such as the information gain ratio or Gini index, RF can be utilized to derive the importance score of each feature in the process of stock movement

prediction [25]. In this work, we optimize the combination of input variables to improve the predictive performance of the machine learning model based on the ranking of feature importance derived from RF. The detailed stock feature selection steps are shown below.

1. First, we choose the mean squared error (MSE) as the criterion for feature selection to build the regression tree. The impurity $Imp(v)$ at node v is shown in Equ. (2):

$$Imp(v) = \sum_{i=1}^O (y_i - y'_i)^2, \quad (2)$$

where y_i and y'_i denote the actual and predicted values of the closing price of the i th sample on node v , and O denotes the number of samples on node v .

2. Then, the impurity change $\Delta_{\mathcal{I}}(v, \mathbf{x}^i)$ of node v after splitting according to the stock feature $\mathbf{x}^i$ is calculated as shown in Equ. (3). Generally, the best-split features will reduce the impurities of node v the most.

$$\Delta_{\mathcal{I}}(v, \mathbf{x}^i) = Imp(v) - P_L * Imp(v^L, \mathbf{x}^i) - P_R * Imp(v^R, \mathbf{x}^i), \quad (3)$$

where $Imp(v^L, \mathbf{x}^i)$ and $Imp(v^R, \mathbf{x}^i)$ denote the impurity of the left child node v^L and the right child node v^R after splitting according to the feature $\mathbf{x}^i$, respectively. P_L denotes the proportion of the number of samples divided from node v to the left child node and the number of samples on node v . P_R denotes the proportion of the number of samples divided from node v to the right child node and the number of samples on node v .

3. Next, the importance score FIM_i^j of the feature $\mathbf{x}^i$ on the j th tree is derived from the impurity change measure $\Delta_{\mathcal{I}}(v, \mathbf{x}^i)$ and normalized to generate $NormFIM_i^j$. The formulas are shown in Equ. (4) and (5):

$$FIM_i^j = \frac{\sum_{b \in V_{\mathbf{x}^i}^j} \Delta_{\mathcal{I}}(b, \mathbf{x}^i)}{\Delta_{\mathcal{I}}(V^j)}, \quad (4)$$

$$NormFIM_i^j = \frac{FIM_i^j}{\sum_{k=1}^K FIM_k^j}, \quad (5)$$

where V^j represents the set of split nodes of the j -tree. $\Delta_{\mathcal{I}}(V^j)$ represents the sum of the impurity changes of all nodes on the j -tree. $b \in V_{\mathbf{x}^i}^j$ represents the nodes containing the feature $\mathbf{x}^i$ in the j -tree and K represents the number of features on the j -tree.

4. Then, the importance scores of the features $\mathbf{x}^i$ on each tree are integrated to generate FIM_i . The formula is shown in Equ. (6):

$$FIM_i = \frac{1}{T} \sum_{j=1}^T NormFIM_i^j, \quad (6)$$

where T is the number of regression trees.

- Finally, we rank the features based on the importance score FIM_i . The subset of features is generated by removing features in order from the bottom of the ranking list. The optimal feature subset is filtered based on the prediction results of the training set.

The stock feature sequence $\mathbf{X} \in \mathbb{R}^{T \times d}$ is filtered through a random forest to generate a new feature sequence $\mathbf{Z} \in \mathbb{R}^{T \times d'}$, where $d' \in \{1, \dots, d-1\}$. The sequence $\mathbf{Z}$ is preprocessed by sliding window and normalization techniques to generate the input samples set $\{\mathbf{Z}_s, \mathbf{Z}_{s+1}, \dots, \mathbf{Z}_T\}$. In the next section, we take the first input sample $\mathbf{Z}_s = (\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{d'}) = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_s)^\top \in \mathbb{R}^{s \times d'}$ as an example to describe the time series decomposition module.

3.3. Moving Average Filter

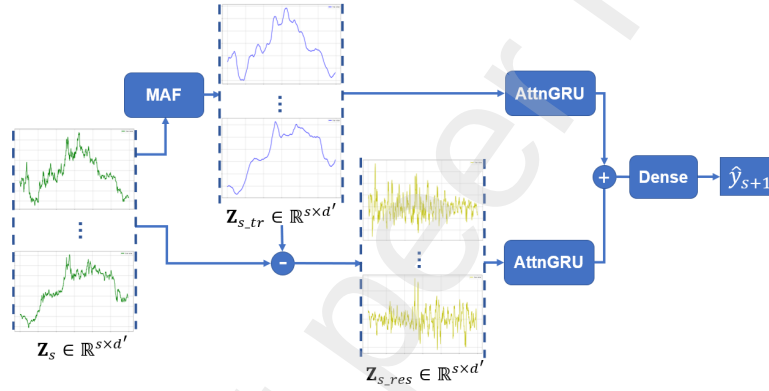


Figure 2: Time series decomposition module based on MAF

In dealing with nonlinear and highly noisy financial time series, a properly decomposed algorithm can reduce modeling difficulties and improve prediction performance. However, time series decomposition algorithms in some studies involve the use of future data in decomposing the test dataset, which leads to inappropriately high prediction accuracy [26]. In this work, to ensure that the decomposition of the time series can only use data from the current moment onwards, we decompose the financial time series by moving average filtering (MAF) to obtain a steady subsequence. The i th feature series of the stock $\mathbf{z}^i$ is used to generate the trend series $\mathbf{z}_{tr}^i \in \mathbb{R}^s$ by MAF. The formula is shown in Equ. (7):

$$\mathbf{z}_{tr}^i = \frac{1}{k} \sum_{j=-q}^q z_{t+j}^i, \quad (7)$$

where k denotes the average pooling size, $q = \lfloor k/2 \rfloor$, $t > q$.

The time series decomposition module is shown in Figure 2. The process of generating $\mathbf{Z}_s$ through MAF for $\mathbf{Z}_{s,tr}$ is divided into two steps. First, we use `Padding1D(.)` to pad the beginning and end of the original input sequence $\mathbf{Z}_s$ to ensure that the average pooled sequence length s remains unchanged. Next, the trend series $\mathbf{Z}_{s,tr}$ is generated using the

average pooling $\text{AvgPool1D}(\cdot)$. Finally, the residual sequence $\mathbf{Z}_{s_tr}$ is extracted from the original input sequence $\mathbf{Z}_s$ by $\mathbf{Z}_{s_res}$. $\mathbf{Z}_{s_tr} \in \mathbb{R}^{s \times d'}$ and $\mathbf{Z}_{s_res} \in \mathbb{R}^{s \times d'}$ are calculated from Equ. (8):

$$\begin{aligned}\mathbf{Z}_{s_tr} &= \text{AvgPool1D}(\text{Padding1D}(\mathbf{Z}_s)), \\ \mathbf{Z}_{s_res} &= \mathbf{Z}_s - \mathbf{Z}_{s_tr}.\end{aligned}\tag{8}$$

Our decomposition algorithm acts on sample sequences with a time window size of s , rather than being applied directly to the entire dataset, which achieves real-time rolling forecasts. These two sequences are used as inputs to the AttnGRU model. Finally, the output of the two AttnGRUs is summed and the predictions are generated via a fully connected layer. Since the trend sequence $\mathbf{Z}_{s_tr}$ and the residual sequence $\mathbf{Z}_{s_res}$ are consistent in the training process in the AttnGRU model, for the sake of brevity, we choose one of the sequences, i.e. the trend sequence $\mathbf{Z}_{s_tr} = (\mathbf{z}_{tr}^1, \mathbf{z}_{tr}^2, \dots, \mathbf{z}_{tr}^{d'}) = (\mathbf{z}_{1_tr}, \mathbf{z}_{2_tr}, \dots, \mathbf{z}_{s_tr})^\top \in \mathbb{R}^{s \times d'}$, as the input sequence of AttnGRU in the following to describe the attention module.

3.4. AttnGRU

3.4.1. Attention

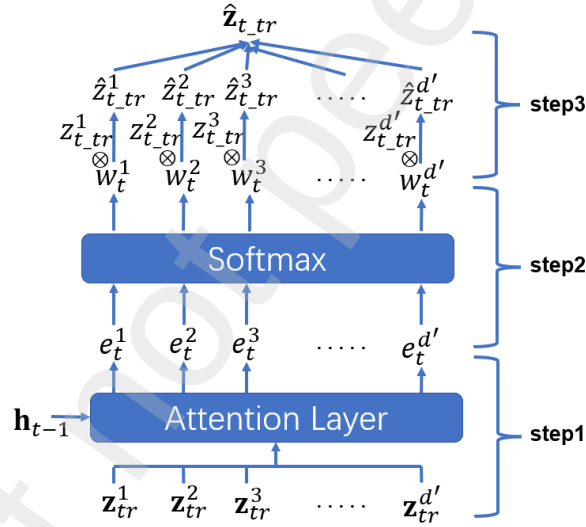


Figure 3: Attention Allocation for Stock Feature Series

Attention is a distribution mechanism, whose core idea is to highlight certain important features of the target. Stock features collected in the financial markets have varying degrees of influence on the closing price and we need to give larger weight to key elements. Inspired by the above, we use the attention mechanism to increase the attention of important features and decrease the attention of redundant features. The main approach is to distribute different weights w to individual features of the input sequence to highlight the importance of certain features.

For a trend sequence $\mathbf{z}_{t_tr}$ at time t , the process of generating an input trend sequence $\hat{\mathbf{z}}_{t_tr} = (\hat{z}_{t_tr}^1, \hat{z}_{t_tr}^2, \dots, \hat{z}_{t_tr}^{d'})^\top \in \mathbb{R}^{d'}$ at time t based on the attention mechanism is shown in

Figure 3. Firstly, the similarity e_t^k between the k th feature trend sequence $\mathbf{z}_{tr}^k \in \mathbb{R}^s$ and the hidden layer state $\mathbf{h}_{t-1} \in \mathbb{R}^m$ at time $t - 1$ in the GRU network is calculated by Equ. (9), where m denotes the number of hidden layer neurons. The vectors $\mathbf{v} \in \mathbb{R}^s$, $\mathbf{W} \in \mathbb{R}^{s \times m}$ and $\mathbf{U} \in \mathbb{R}^{s \times s}$ are the parameters that the model needs to learn.

$$e_t^k = \mathbf{v}^T \tanh(\mathbf{W}\mathbf{h}_{t-1} + \mathbf{U}\mathbf{z}_{tr}^k), k = 1, \dots, d'. \quad (9)$$

Secondly, the similarity e_t^k is normalized to the attention weight w_t^k through the softmax layer. The formula is shown in Equ. (10):

$$w_t^k = \text{softmax}(e_t^k) = \frac{\exp(e_t^k)}{\sum_{i=1}^{d'} \exp(e_t^i)}. \quad (10)$$

Finally, each feature in the input sequence is multiplied by the corresponding attention weight respectively. The trend input sequence $\hat{\mathbf{z}}_{t,tr}$ at time t based on the attention mechanism is shown in Equ. (11):

$$\hat{\mathbf{z}}_{t,tr} = (w_t^1 z_{t,tr}^1, w_t^2 z_{t,tr}^2, \dots, w_t^{d'} z_{t,tr}^{d'})^\top. \quad (11)$$

3.4.2. GRU

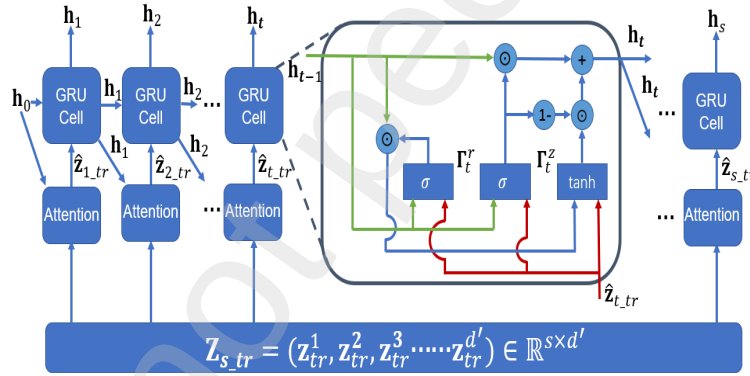


Figure 4: Illustration of the architecture of the AttnGRU

Recurrent neural networks (RNNs) are widely used to analyze and predict time series trends. The GRU includes an update gate and a reset gate. The update gate is used to control how much information from the previous time state is brought into the current state. The higher the value of the update gate, the more information is brought into the current state. The reset gate is designed to control how much information from the previous state needs to be forgotten. Conversely, the lower the value of the reset gate, the more information is ignored. Figure 4 shows the structure of the GRU network based on the attention mechanism. Firstly, the update gate state Γ_t^z and the reset gate state Γ_t^r are established by the input $\hat{\mathbf{z}}_{t,tr} \in \mathbb{R}^{d'}$ at time t and the hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^m$ at time $t - 1$. The formula is shown in Equ. (12):

$$\begin{aligned} \Gamma_t^z &= \sigma(\mathbf{W}_z \cdot [\mathbf{h}_{t-1}, \hat{\mathbf{z}}_{t,tr}]) \\ \Gamma_t^r &= \sigma(\mathbf{W}_r \cdot [\mathbf{h}_{t-1}, \hat{\mathbf{z}}_{t,tr}]). \end{aligned} \quad (12)$$

Next, the candidate hidden state $\tilde{\mathbf{h}}_t$ for time t is derived from Equ. (13):

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \cdot [\mathbf{\Gamma}_t^r \odot \mathbf{h}_{t-1}, \hat{\mathbf{z}}_{t.tr}]). \quad (13)$$

Finally, the candidate hidden state $\tilde{\mathbf{h}}_t$ at time t and the hidden state $\mathbf{h}_{t-1}$ at time $t-1$ are linearly combined to generate the hidden node output $\mathbf{h}_t$ at the current moment shown in Equ. (14):

$$\mathbf{h}_t = \mathbf{\Gamma}_t^z \odot \mathbf{h}_{t-1} + (1 - \mathbf{\Gamma}_t^z) \odot \tilde{\mathbf{h}}_t, \quad (14)$$

where $\mathbf{W}_z$, $\mathbf{W}_r$ and $\mathbf{W}_h \in \mathbb{R}^{m \times d'}$ represent the weight matrices which the model is to learn. σ is the sigmoid function, and $\odot$ denotes the Hadamard product. The AttnGRU outputs of the two sub-series are summed and the predictions are generated through a fully connected layer.

4. Experimental setup

We develop trading strategies based on the prediction results to further illustrate the usefulness of the proposed model. The specific settings of the trading strategy are described in this section. Firstly, we present the stock data for companies listed in China or the US; Then, the setup of the comparison experiment is described; Finally, we show the detailed setup of the trading strategy.

4.1. Dataset

In this work, we select six stocks in different countries or regions of the stock market as illustrative examples. Historical data is collected from the East Money website¹ and the API interface of BaoStock². The subjects in the Chinese A-share market are BOE (sz.000725) and Silan Microelectronics (sh.600460). In the Hong Kong H-share market, the subjects were Hong Kong Exchanges and Clearing Limited (00388) and Build Your Dreams (01211). The US stock market experiments involved Qualcomm (QCOM) and Texas Instruments (TXN). The research data spans the period from June 2010 to June 2022, including approximately 3000 tradings. The most dominant technical indicators in stock trading include trend indicators, momentum indicators and volume indicators. Therefore, in addition to the raw stock data, we select three different types of technical indicators (trend, momentum, and volume) as input features [27]. The feature variables collected, including their formulas or descriptions, are summarized in Table 1. The Chinese A-share market, mainly dominated by individual investors, is prone to herd behaviour. Therefore, we introduce market indexes, i.e. the Shenzhen Securities Component Index (SZI) or Shanghai Securities Composite Index (SSEC), into the input characteristics of the experimental stocks in the Chinese A-share market [28]. As the range of different input features varies inconsistently, we scaled the

¹www.eastmoney.com

²www.baostock.com

Table 1: Features for stock price forecasting models

No	Feature	Description or Formula
x_t^1	C_t	Closing price at time t
x_t^2	H_t	Highest price at time t
x_t^3	L_t	Lowest price at time t
x_t^4	O_t	Opening price at time t
x_t^5	$\%CHg_t$	$\frac{C_t - C_{t-1}}{C_{t-1}} \times 100$
x_t^6	$\%T_t$	Turnover rate at time t
x_t^7	V_t	Volume at time t
x_t^8	T_t	Turnover at time t
x_t^9	$C(s)_t$	SSEC or SZI closing price at time t
x_t^{10}	$H(s)_t$	SSEC or SZI highest price at time t
x_t^{11}	$L(s)_t$	SSEC or SZI lowest price at time t
x_t^{12}	$O(s)_t$	SSEC or SZI opening price at time t
x_t^{13}	$PC(s)_t$	SSEC or SZI previous closing price at time t
x_t^{14}	$CHg(s)_t$	$C(s)_t - C(s)_{t-1}$
x_t^{15}	$\%CHg(s)_t$	$\frac{C(s)_t - C(s)_{t-1}}{C(s)_{t-1}} \times 100$
x_t^{16}	$V(s)_t$	SSEC or SZI Volume of the listed company at time t
x_t^{17}	$T(s)_t$	SSEC or SZI Turnover of the listed company at time t
x_t^{18}	$MA12_t$	$\frac{1}{12} \sum_{i=0}^{n-1} C_{t-i}$, where $n = 12$
x_t^{19}	$MA24_t$	$\frac{1}{24} \sum_{i=0}^{n-1} C_{t-i}$, where $n = 24$
x_t^{20}	$EMA12_t$	$\alpha C_t + (1 - \alpha) EMA12_{t-1}$, where $\alpha = \frac{2}{13}$
x_t^{21}	$EMA26_t$	$\alpha C_t + (1 - \alpha) EMA26_{t-1}$, where $\alpha = \frac{2}{27}$
x_t^{22}	$MOM12_t$	$C_t - C_{t-n+1}$, where $n = 12$
x_t^{23}	$MOM24_t$	$C_t - C_{t-n+1}$, where $n = 24$
x_t^{24}	DIF_t	$EMA12_t - EMA26_t$
x_t^{25}	OBV_t	$OBV_{t-1} + \begin{cases} V_t & \text{if } C_t > C_{t-1} \\ 0 & \text{if } C_t = C_{t-1} \\ -V_t & \text{if } C_t < C_{t-1} \end{cases}$
x_t^{26}	1RSI_t	$100 - \frac{100}{1 + \frac{sum_up(n)}{sum_dw(n)}}$, where $n = 14$
x_t^{27}	2CII_t	$\frac{M_t - MA_t}{0.015 D_t}$

¹ $sum_up(n)$ represents the sum of the upward price change over the last n days, $sum_dw(n)$ represents the sum of the downward price change over the last n days.

² $M_t = (C_t + H_t + L_t)/3$, $MA_t = \sum_{i=0}^{n-1} M_{t-i}/n$, $D_t = \sum_{i=0}^{n-1} |M_{t-i} - MA_t|/n$

dataset to the range $[0,1]$ by the Min-Max normalization technique in order to accelerate the convergence of the deep learning model. The z_{norm} formula is shown in Equ. (15):

$$z_{norm} = \frac{z - z_{min}}{z_{max} - z_{min}}, \quad (15)$$

where z_{min} and z_{max} are the minimum and maximum values in the training dataset, respectively. The entire dataset is divided into two parts, with 80 % of the dataset serving as the training dataset and 20 % of the remaining dataset acting as the test dataset.

4.2. Comparison of experimental setup

Five machine learning models were selected to evaluate the performance of the proposed models: CNN, LSTM, GRU, support vector regression (SVR) and BLS (Broad Learning System). In addition, time series forecasting models constructed by researchers in recent years, such as SCINet [29], Dlieanr [30], TCN [31] and DA-RNN [32], were also added to the comparison experiments. To measure the difference between actual values and predicted values, we used four performance indicators in our experiments: mean square error (MSE), mean absolute error (MAE), mean absolute percentage error ($MAPE$), and coefficient of determination (R^2). These indicators are mathematically defined as shown in Equ. (16):

$$\begin{aligned} MSE &= \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2, \\ MAE &= \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|, \\ MAPE &= \frac{100\%}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|, \\ R^2 &= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \end{aligned} \quad (16)$$

where n denotes the number of samples in the test set, y_i and $\hat{y}_i$ represent the actual and predicted values, respectively. $\bar{y}$ is the average of the closing prices of the n samples. For better prediction performance, MSE , MAE and $MAPE$ should be close to zero, while R^2 should be close to one.

In addition to evaluating the accuracy of the predictive models, we also verified whether there were statistically significant differences in the performance of the different models. The nonparametric recommendation statistical test recommended by Demvar [33], which includes the Friedman test [34] and the Iman-Davenport test [35], is used in this work. The test evaluation process consists of two steps: 1) a combination of different evaluation metrics is used to rank the performance of all the algorithms to obtain an average performance ranking for each algorithm; 2) the Friedman test and the Iman-Davenport test are applied to verify whether all the algorithms perform equally.

In Step one, we rank the predictive performance of the ten algorithms by combining four evaluation metrics. The algorithm with the best performance is ranked one, then

the algorithm with the second-best performance is ranked two, and so on. In the case of equal performance, the rankings will be equally distributed. The formula for calculating the average algorithm performance ranking R_j is shown in Equ. (17):

$$R_j = \frac{1}{W} \frac{1}{I} \sum_{w=1}^W \sum_{i=1}^I r_i^{j,w}, \quad (17)$$

where $r_i^{j,w}$ denotes the w th indicator ranking on the j th algorithm in the i th dataset, W is the number of indicators, and I indicates the number of datasets.

In step two, we determine whether there is a significant difference in performance between the different algorithms based on the Friedman and Iman-Davenport test. The null hypothesis H_0 in the test is that all algorithms have the same predictive performance. The formula for calculating the Friedman test based on the average ranking of the algorithm performance R_j is shown in Equ. (18):

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right], \quad (18)$$

where k denotes the number of algorithms and N denotes the number of experiments. The statistic χ_F^2 follows a χ^2 distribution with $k-1$ degrees of freedom. Iman and Davenport consider the Friedman test to be conservative and propose the statistic F_F , which follows an F distribution with $k-1$ and $(k-1) \times (N-1)$ degrees of freedom. The Iman-Davenport test based on the statistic χ_F^2 is derived by Equ. (19):

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1) - \chi_F^2}. \quad (19)$$

In the model comparison experiment, we evaluate the performance of the proposed model in predicting the closing price of stocks utilizing four regressors and hypothesis tests based on the ranking of the indicators.

4.3. Algorithmic Trading

Table 2: Backtest parameter settings

Parameter	Description or Formula	Value
TP	Transaction Period	03/10/2021-06/30/2022
IC	Initial Capital	300000
TF	Transaction Fee	0.005
OpenPerc	Ratio of order amount bought to funds held	0.99
$z(\%)$	$\frac{\hat{y}_{t+1} - O(t+1)}{O(t+1)} \times 100$	0.2

According to the prediction results of the proposed model, we can set up trading strategies. There are three types of stock trading day operations: buying, selling and holding.

Based on the relationship between the predicted value of the closing price $\hat{y}_{t+1}$ on the day $t + 1$ and the true value y on the day t , the following two trading cases are given.

1. If $\hat{y}_{t+1}$ is higher than y , the share price is rising in the future. Setting up a buy if the stock is not held on the day t ; No operation if the stock is held on the day t .
2. If $\hat{y}_{t+1}$ is lower than y , the share price is decreasing in the future. Setting up a sale if the stock is held on the day t ; No operation if the stock is not held on the day t .

In addition, according to stock market-related knowledge, the closing price on the day is not consistent with the opening price of the next day. When we set up a buy or sell based on the day's closing price, the stock market is already closed. To simulate a real trading scenario, the opening price of the next day is used as the price per share of the trade order. In brief, when buying or selling operation is set up on the day, we will execute the trade order at the opening of the stock market the next day. Secondly, there are fees for actual stock market transactions. To reduce the increase in transaction costs due to frequent trading, we set a psychological threshold z for investors. Only if the predicted value of the closing price on the day $t + 1$ is greater than the actual value on the day t and also higher than the opening price $z\%$ on the day $t + 1$, the stock will be bought at the opening on the day $t + 1$. In conclusion, the opening price of the next day is used as the price per share of the trade order.

In order to compare the return potential of the proposed model under actual stock trading, three traditional trading strategies are adopted: Buy & Hold (B&H) [11], Relative Strength Index(RSI) [36] and Moving Average Convergence/Divergence(MACD) [37].

- In the B&H strategy, each stock is bought at the start of the backtesting period and sold on the last day of the backtesting period without trading halfway through.
- The RSI indicator can infer trading momentum and identify general trends. The formula is shown in Table 1. The RSI value ranges from 0 to 100. When the RSI value is below 30, the asset is considered oversold and a buy operation is executed at this point. When the RSI value is above 70, the asset is considered overbought and a sell operation is executed. No trades are executed on trading days when the RSI value is not in the mentioned range.
- The MACD indicator has evolved from the Exponential Moving Average (EMA). The indicator consists of the DIF, DEA and MACD histogram. The EMA and DIF formulas are shown in Table 1. DEA is calculated from the 9-day EMA of the DIF. MACD Strategy is generated based on the crossover of DIF and DEA. When DIF and DEA are above the horizontal axis, a buy operation is executed if the DIF rises to intersect the DEA. When DIF and DEA are below the horizontal axis, a sell operation is executed if DEA falls to intersect DIF. No trades are executed on trading days when DEA and DIF do not satisfy the above intersection conditions.

The parameters for the backtesting period are shown in Table 2. The proposed model is compared with the three traditional trading strategies in terms of cumulative returns (CR),

return on investment (ROI), and annual return (AR). ROI and AR are shown in Equ. (20):

$$\begin{aligned} ROI &= \frac{\text{Cumulative Returns}}{\text{Initial Capital}} \times 100 \\ AR &= \left[\left(1 + \frac{\text{Cumulative Returns}}{\text{Initial Capital}} \right)^{\frac{365}{D}} - 1 \right] \times 100, \end{aligned} \quad (20)$$

where D represents the number of days of investment.

5. Experiment result

After describing the experimental setup, we illustrate the results of the prediction experiment, the ablation experiment and the backtest.

5.1. Comparative performance evaluation

Table 3: Feature importance rank

	000725	600460	00388	001211	QCOM	TXN
C_t	1	1	1	1	1	1
H_t	2	2	4	4	3	4
L_t	3	3	2	2	2	2
O_t	4	4	3	3	4	3
$\%CHg_t$	26	24	16	16	12	16
$\%T_t$	14	18	17	14	11	17
V_t	11	17	10	10	17	12
T_t	9	9	14	13	10	15
$C(s)_t$	22	14	-	-	-	-
$H(s)_t$	25	12	-	-	-	-
$L(s)_t$	19	11	-	-	-	-
$O(s)_t$	23	15	-	-	-	-
$PC(s)_t$	21	13	-	-	-	-
$CHg(s)_t$	18	27	-	-	-	-
$\%CHg(s)_t$	27	25	-	-	-	-
$V(s)_t$	13	20	-	-	-	-
$T(s)_t$	15	16	-	-	-	-
$MA12_t$	6	6	6	6	6	6
$MA24_t$	8	8	8	8	8	8
$EMA12_t$	5	5	5	5	5	5
$EMA26_t$	7	7	7	7	7	7
$MOM12_t$	16	21	13	18	12	10
$MOM24_t$	10	19	11	11	14	13
DIF_t	24	23	12	12	16	11
OBV_t	17	22	15	15	13	14
RSI_t	20	26	18	17	15	18
CII_t	12	10	9	9	9	9

The results of the stock feature importance ranking are shown in Table 3, where the feature with the highest importance score is ranked 1. The symbol "-" indicates that the

feature is not present in Hong Kong and US stocks. Tables 4, 5, 6, 7, 8 and 9 present the results of the experiment for the six stocks. The bold numbers indicate the best result, while underlined numbers indicate the second-best performance. The experimental results show that our proposed model is the best in each regression metric among the six experimental stocks, achieving the lowest forecast error (MSE , MAE , and $MAPE$) and the highest forecast accuracy (R^2). The prediction results of the proposed model are closer to the actual price compared to other single models. CNN shows good prediction results in most of the experimental stocks without considering the proposed model. Several state-of-the-art time series forecasting models outperform classical neural network models for specific stocks. DA-RNN has R^2 of 0.9749 in QCOM and SCINet has R^2 of 0.9828 in 01211, second only to our proposed model. In addition, Figure 5 represents the fit results between the predicted and true values of the closing price for the experimental stocks on the training and test sets.

In the Friedman and Iman-Davenport test, we set $k=10$ and $N = 4 \times 6 = 24$. The average ranking of the ten models in the six datasets for the four indicators is shown in Table 10. According to Eqs. (18) and (19), $\chi_F^2=45.15$ and $F_F=6.07$ are calculated. As shown in Table 11, χ_F^2 is greater than the critical value of 16.919 for the χ^2 distribution with $9=(10-1)$ degrees of freedom. F_F is greater than the critical value of 2.32 for the F distribution with $9=(10-1)$ and $9 \times 23=(4 \times 6-1)$ degrees of freedom. Therefore, according to this result, we reject the hypothesis that the 10 algorithms have the same predictive performance. The results of the model comparison experiments show that our proposed model exhibits the best performance in predicting stock prices.

Table 10: Average ranking of each indicator for ten models

Model	MSE	MAE	MAPE	R^2	Avg.rank
BLS	7.00	6.50	6.33	6.67	6.63
SVR	7.83	8.50	8.83	7.83	8.25
CNN	<u>3.00</u>	<u>3.17</u>	<u>3.17</u>	<u>3.50</u>	<u>3.21</u>
GRU	6.17	6.17	5.67	5.67	5.92
LSTM	6.67	6.67	6.67	7.33	6.83
DA-RNN	5.83	5.50	5.33	5.00	5.42
Dlinear	8.00	8.17	8.50	8.17	8.21
SICNet	4.33	4.00	4.50	4.33	4.29
TCN	5.17	5.33	5.00	5.50	5.25
RF-MAF-AttnGRU	1.00	1.00	1.00	1.00	1.00

Table 11: Friedman and Iman-Davenport tests results for ten models

Method	Test values	Distribution values	Hypothesis
Friedman	45.15	16.919	Reject
Iman-Davenport	6.07	2.32	Reject

Table 4: 00388: The evaluation value of different models

model	MSE	MAE	MAPE	R^2
BLS	1012.9941	20.4763	4.8839	0.8666
SVR	437.3241	17.0990	4.1177	0.9424
CNN	187.4128	9.7891	2.4949	0.9738
GRU	354.4064	14.7777	3.5226	0.9483
LSTM	552.7646	17.5699	4.1720	0.9194
DA-RNN	206.6058	10.6797	2.6404	0.9728
Dlinear	737.7153	21.4995	5.0184	0.8969
SICNet	199.4865	10.6783	2.7328	0.9721
TCN	283.6471	12.8388	3.1114	0.9475
RF-MAF-AttnGRU	151.8659	8.7636	2.2262	0.9794

Table 6: 01211: The evaluation value of different models

model	MSE	MAE	MAPE	R^2
BLS	450.6806	15.3495	8.0597	0.9406
SVR	741.2957	22.3738	11.1777	0.9023
CNN	122.5417	8.0780	4.4357	0.9825
GRU	862.6669	22.9845	10.2985	0.8767
LSTM	1005.8549	24.1866	10.7843	0.8602
DA-RNN	151.1044	8.9145	5.0573	0.9799
Dlinear	1874.1820	32.8133	14.9387	0.7395
SICNet	123.8531	8.1985	4.5882	0.9828
TCN	130.5309	8.3631	4.6843	0.9813
RF-MAF-AttnGRU	116.5469	7.8717	4.4188	0.9842

Table 8: QCOM: The evaluation value of different models

model	MSE	MAE	MAPE	R^2
BLS	65.0215	5.6291	4.0951	0.9351
SVR	279.0380	14.8639	10.7945	0.7217
CNN	24.4098	3.5810	2.7465	0.9719
GRU	45.8520	5.0059	3.5947	0.9472
LSTM	45.9267	5.1147	3.8668	0.9471
DA-RNN	25.1617	3.6052	2.8343	0.9749
Dlinear	82.5624	6.2390	4.5536	0.9122
SICNet	30.5397	4.0320	3.1239	0.9675
TCN	27.0822	3.6604	2.8711	0.9712
RF-MAF-AttnGRU	22.9047	3.3827	2.6860	0.9769

Table 5: 00725: The evaluation value of different models

model	MSE	MAE	MAPE	R^2
BLS	0.0250	0.1182	2.4182	0.9630
SVR	0.0261	0.1314	2.8388	0.9613
CNN	0.0274	0.1249	2.5755	0.9604
GRU	0.0251	0.1184	2.4420	0.9636
LSTM	0.0248	0.1181	2.4383	0.9642
DA-RNN	0.0295	0.1245	2.5436	0.9564
Dlinear	0.0259	0.1196	2.4745	0.9625
SICNet	0.0269	0.1196	2.4699	0.9611
TCN	0.0290	0.1245	2.5344	0.9587
RF-MAF-AttnGRU	0.0226	0.1110	2.2793	0.9684

Table 7: 600460: The evaluation value of different models

model	MSE	MAE	MAPE	R^2
BLS	10.4951	2.0770	4.8762	0.9684
SVR	20.8829	3.2801	7.2322	0.9372
CNN	5.5099	1.6704	4.7560	0.9834
GRU	31.6111	3.9320	8.4345	0.9047
LSTM	29.4950	3.8186	8.4345	0.9104
DA-RNN	7.6436	1.8584	4.8777	0.9770
Dlinear	27.6315	3.5334	8.7171	0.9167
SICNet	6.6589	1.7993	4.9402	0.9799
TCN	6.1929	1.7643	4.8044	0.9813
RF-MAF-AttnGRU	3.8322	1.3727	3.8799	0.9885

Table 9: TXN: The evaluation value of different models

model	MSE	MAE	MAPE	R^2
BLS	32.7355	4.4056	2.6261	0.9139
SVR	40.1551	5.3998	3.0962	0.8577
CNN	17.3815	3.2326	1.9052	0.9384
GRU	18.5325	3.4892	2.0400	0.9400
LSTM	26.9288	4.2807	2.4725	0.9046
DA-RNN	29.1912	4.5064	2.6176	0.9222
Dlinear	28.7135	4.3673	2.5106	0.9071
SICNet	20.6155	3.5335	2.0950	0.9333
TCN	28.6525	4.4236	2.5820	0.9073
RF-MAF-AttnGRU	14.6825	2.9944	1.7828	0.9592

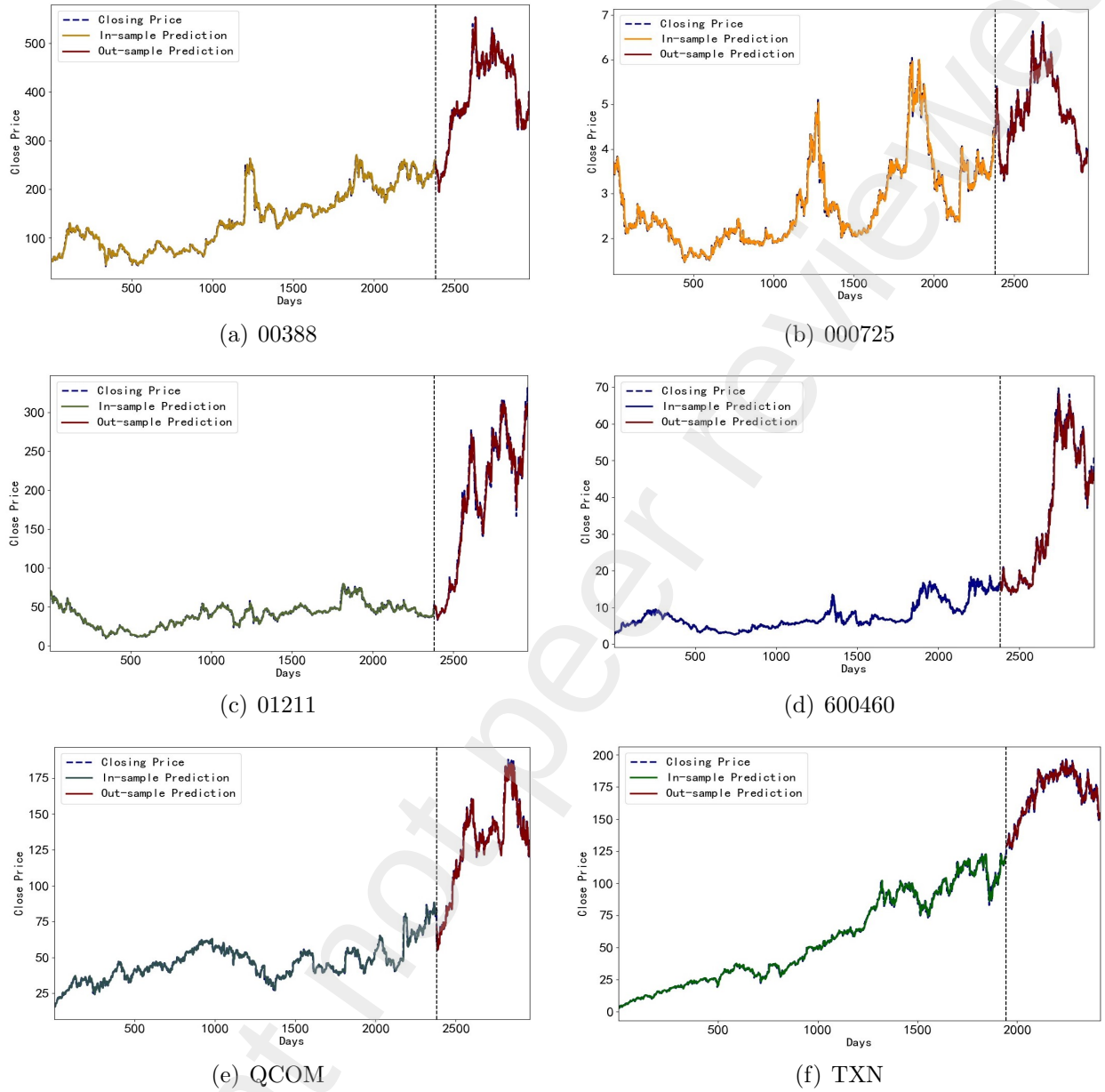


Figure 5: Prediction results of RF-MAF-AttnGRU on the training and test sets of six experimental stocks; (a) 00388; (b) 000725; (c) 01211; (d) 600460; (e) QCOM; (f) TXN.

5.2. Ablation study

The proposed model contains three modules: the feature selection module, the time series decomposition module, and the attention mechanism module. To further evaluate the effectiveness of each module used in the RF-MAF-AttnGRU, we conducted ablation experiments on six stocks. The model variations are shown below:

- w/o MAF (Moving Average Filter) is a variation of the full RF-MAF-AttnGRU without the time series decomposition module. The time series selected by the RF is not

decomposed into trend series or residual series and are fed directly into the AttnGRU model.

- w/o Attn (Attention mechanism) is a variation of the full RF-MAF-AttnGRU without the attention mechanism module. The input time series are not feature weighted by the attention mechanism and are fed directly into the GRU model.
- w/o RF(Random Forest) is a variation of the full RF-MAF-AttnGRU without the feature selection module. Instead, we use all of the collected features, namely, without feature selection.
- Full Model is the complete model of the component.

Table 12: Ablation experiments with the RF-MAF-AttnGRU model on six experimental stock datasets

model	000388				000725			
	MSE	MAE	MAPE	R^2	MSE	MAE	MAPE	R^2
w/o MA	161.8005	9.1422	2.3215	0.9774	0.0248	<u>0.1131</u>	<u>2.3243</u>	0.9647
w/o Attn	<u>156.5095</u>	<u>8.9615</u>	<u>2.2652</u>	<u>0.9781</u>	<u>0.0240</u>	0.1157	2.3890	<u>0.9653</u>
w/o RF	397.5360	15.7630	3.8172	0.9445	0.0291	0.1321	2.7438	0.9579
Full Model	151.8659	8.7636	2.2262	0.9794	0.0226	0.1110	2.2793	0.9684
model	001211				600460			
	MSE	MAE	MAPE	R^2	MSE	MAE	MAPE	R^2
w/o MA	<u>121.3842</u>	<u>8.0438</u>	<u>4.4677</u>	<u>0.9831</u>	<u>5.0609</u>	<u>1.5454</u>	<u>4.1234</u>	<u>0.9847</u>
w/o Attn	241.5603	11.6812	5.8333	0.9664	6.3078	1.7239	4.4447	0.9810
w/o RF	153.4594	9.1809	5.0464	0.9787	6.9308	1.7913	4.7931	0.9791
Full Model	116.5469	7.8717	4.4188	0.9842	3.8322	1.3727	3.8799	0.9885
model	QCOM				TXN			
	MSE	MAE	MAPE	R^2	MSE	MAE	MAPE	R^2
w/o MA	24.7676	3.5226	2.7299	0.9715	16.7804	3.2110	1.8914	0.9457
w/o Attn	<u>23.5040</u>	<u>3.4040</u>	<u>2.7155</u>	<u>0.9750</u>	<u>15.4063</u>	<u>3.0931</u>	<u>1.8241</u>	<u>0.9501</u>
w/o RF	38.7450	4.5818	3.4897	0.9588	17.2435	3.2829	1.9468	0.9442
Full Model	22.9047	3.3827	2.6860	0.9769	14.6825	2.9944	1.7828	0.9592

Table 12 presents the results of the ablation experiment for the RF-MAF-AttnGRU model. In general, the full model results for each metric are optimal for each stock, and removing any of the modules would reduce the predictive performance.

When the attention mechanism is removed from the full model, the results for each metric show a reduction in prediction accuracy, which demonstrates that the input sequence without the designation of feature weights reduces prediction performance [38]. Therefore, we believe that the introduction of an attention mechanism to give different weights for the input features is necessary. When MAF is removed from the full model, all performance metrics degrade, demonstrating the difficulty of machine learning models to directly model complex

financial time series. Extracting relatively stable trend series and residual sequences from the time series can reduce the difficulty of the forecasting task and improve the forecasting accuracy [9]. When RF is removed from the full model, this leads to a decrease in all aspects of model performance, which demonstrates that feature selection methods can filter redundant features and improve model prediction accuracy to some extent [8, 18]. In a complex stock market, many factors affect the closing price and the extent of their influence varies. As our data collection may be redundant, RFs are used to score the input features and find the optimal subset of features to improve the accuracy of the model. The results of the ablation experiments show that the variant model with the second highest performance includes Random Forest as the feature selection module, which verifies that while each component improves the model prediction performance in different ways, the Random Forest feature selection module is the most critical component of the proposed model.

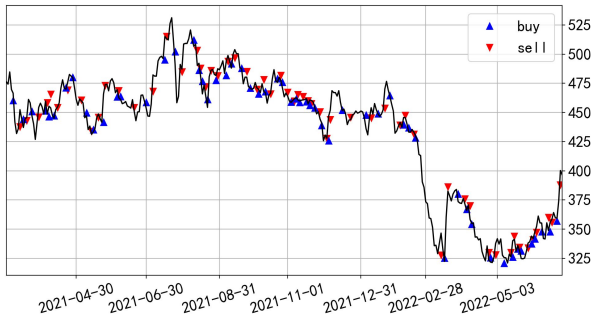
In conclusion, each module of the RF-MAF-AttnGRU model can play a significant role in predicting stock market movements and thus validate the effectiveness of the proposed model design.

5.3. Back-testing results

To compare the proposed model's ability to generate returns in the actual trading scenario, we develop a trading strategy based on the predictions of the proposed model and compare it with three other traditional trading strategies (B&H, MACD, and RSI) in terms of profitability indicators. The buy and sell signals based on the proposed model for the six experimental stocks are shown in Figures 6(a), 6(b), 6(c), 6(d), 6(e), and 6(f). The overall returns of the four trading strategies in the six experimental stocks are shown in Table 14. Over the backtesting interval, the CR, ROI and AR of the proposed model are 94825, 31.608% and 23.443%, respectively. In comparison, the AR of B&H, MACD, and RSI are 8.17%, 4.585%, and -4.96%, respectively. Hence, although the trading strategy based on the proposed model has frequent trading characteristics, which have been taken into account transaction costs, this trading strategy not only achieves positive returns during the backtest period but also generates significantly higher returns than other traditional trading strategies.

Figures 7(a), 7(b), 7(c), 7(d), 7(e), and 7(f) show the specific profits of the six experimental stocks, further illustrating the return performance of our strategy compared to other strategies. Some literature suggests that the models are able to predict accurately when share prices are in a downtrend and underperform when it is in an uptrend [39]. Figure 13 shows the closing prices at the beginning and end of the backtesting phase, the value of the rise and fall (CHg), and the rate of rise and fall (%CHg) for the six experimental stocks. Obviously, there are two scenarios where the experimental stocks have a general trend up or a general trend down. The return results show that our model is able to make valid forecasts and achieve returns in both of these market scenarios. In addition, we introduce maximum retracement (MDD) as a risk indicator in the experimental stocks [11], which shows the maximum loss that the strategy may face during execution. Profitability indicators for the B&H strategy outperformed in the particular experimental stock, which indicates that the share price generally increased during the backtest period. However, the

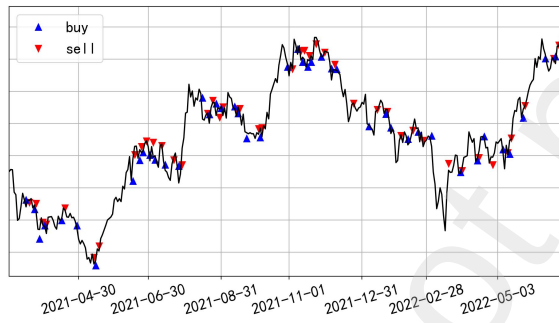
overall maximum drawdown of the B&H strategy is higher than other trading strategies, which is a risk that investors have difficulty undertaking. In contrast, our model achieves the highest profitability with a lower overall maximum drawdown. In summary, in terms of application prospects, our proposed model is able to adapt to complex and volatile stock markets and make accurate predictions. In a trade-off between return and risk, the trading strategies developed by the proposed model are more suitable for investors than traditional trading strategies.



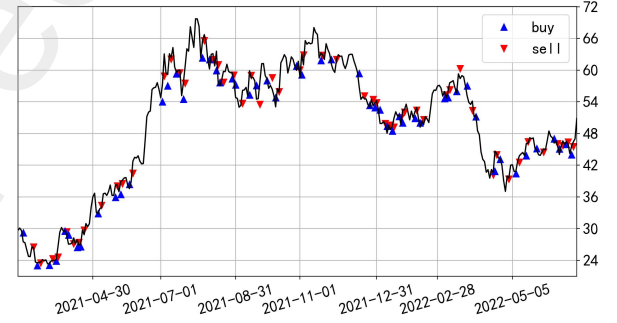
(a) 00388



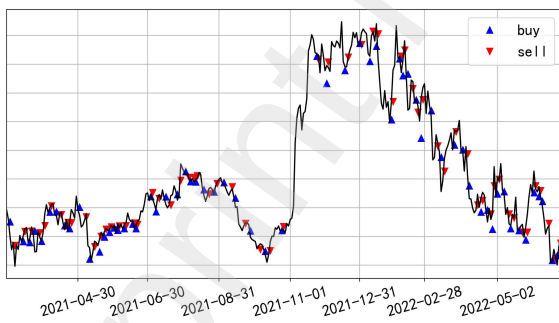
(b) 000725



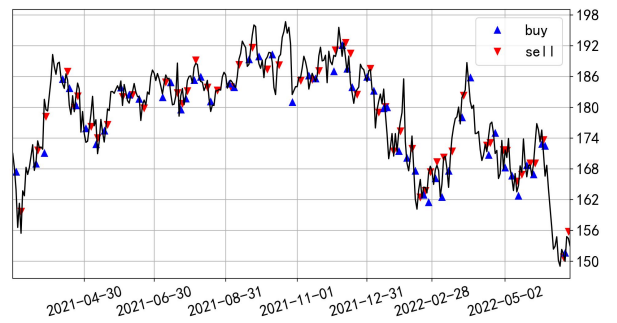
(c) 01211



(d) 600460



(e) QCOM



(f) TXN

Figure 6: Buy and sell signals based on the RF-MAF-AttnGRU model for six experimental stocks; (a) 00388; (b) 000725; (c) 01211; (d) 600460; (e) QCOM; (f) TXN.

Table 13: Description of the closing price of experimental stocks during the backtest

Stock	Start	End	CHg	%CHg
00388	476.27	386	-90.27	-18.95
000725	5.75	3.94	-1.81	-31.48
01211	212.7	314	101.3	47.63
600460	26.69	52	25.31	94.83
QCOM	135.37	127.74	-7.63	-5.64
TXN	171.03	152.5	-18.53	-10.83

Table 14: Overall investment returns of the four trading strategies

Trade Strategy	Initial Capital	Final Capital	CR	ROI(%)	AR(%)
Our Strategy	300000	394825	94825	31.608	23.443
B&H	300000	332352	32352	10.784	8.17
MACD	300000	318062	18062	6.021	4.585
RSI	300000	280743	-19257	-6.419	-4.96

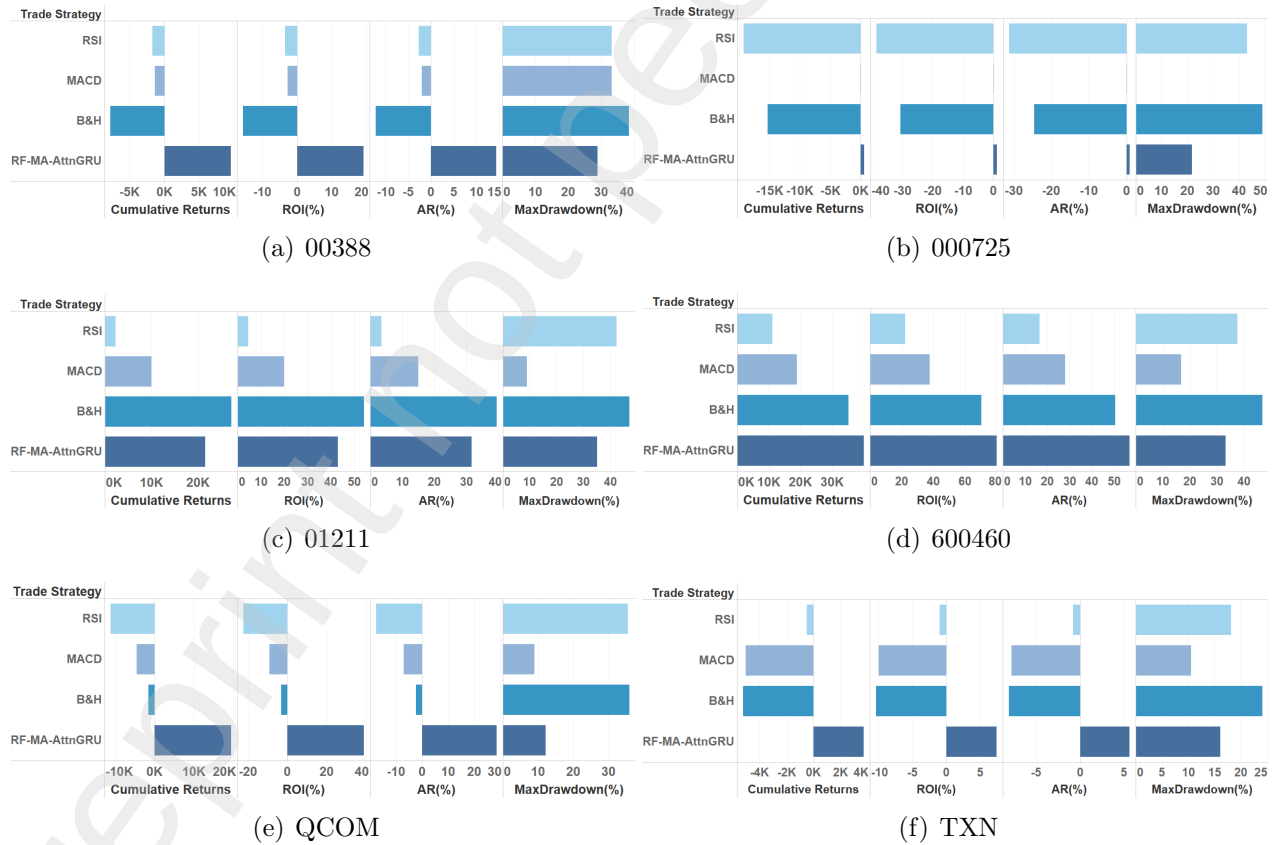


Figure 7: Return results of four trading strategies on six adopted stocks; (a) 00388; (b) 000725; (c) 01211; (d) 600460; (e) QCOM; (f) TXN.

6. Conclusion

The main objective of this work is to develop new model to accurately predict stock price movements and make reasonable trading decisions for investors with less risk. In this study, a hybrid framework called RF-MAF-AttGRU is proposed to predict the price movements of the stock market through the combination of GRUs based on attention mechanism, RF, and MAF. Firstly, the input features, such as raw stock prices and technical indicators, are ranked in importance score by RF. Secondly, we decompose the selected time series into trend and residual series by moving averages. Then, the gated recurrent units based on the attention mechanism are applied to model these two series for the price prediction task. In the model performance comparison experiments, our proposed model showed excellent prediction performance, outperforming other comparison models in all four regression evaluation metrics. In particular, we demonstrate the effectiveness of the feature selection module, the sequence decomposition module, and the input sequence processing module in the performance comparison between the proposed model and its variants.

This is an important part of the process of validating the usefulness of the model, and we develop corresponding trading strategies based on the predictions of the proposed model to help investors facilitate their trading operations. The results of the backtest show that the trading strategy based on the proposed model was profitable in all six experimental stocks, taking into account the commission. The AR of the trading strategy based on the proposed model is 23.44%, compared to the best AR of 8.17% for the traditional trading strategy. The proposed model is the most desirable in terms of both application prospects and the trade-off between risk and return.

Future research can focus on the following aspects: (1) Each module in the model has its parameters and we can explore a parameter tuning method to optimize the model performance for wider and more accurate applications in financial time series forecasting; (2) The adaptability and flexibility of the model are further explored by extending it to other financial domains, such as futures, exchange rates, and cryptocurrencies.

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